

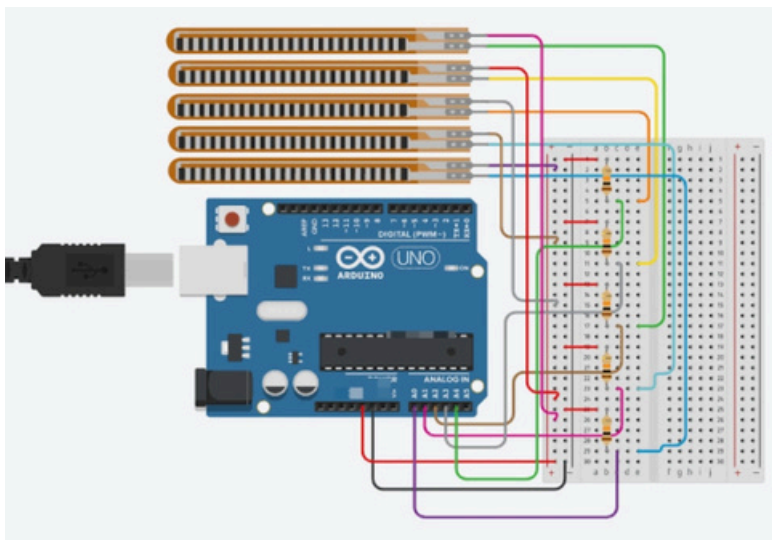
Introduction

This project presents a wearable glove system designed to assist individuals with speech impairments by translating their hand gestures into spoken words. By capturing the natural movements of the hand, the system identifies predefined gestures and converts them into clear, audible speech through a connected device. This enables seamless, real-time communication without relying on sign language expertise from the listener. The solution is portable, intuitive, and suitable for use in daily interactions, education, healthcare, and other environments where clear communication is essential.



Components

- Arduino Nano Every – processes sensor data and recognizes gestures
- Flex Sensors (5x) – detect bending of fingers
- BNO055 IMU – tracks hand motion and orientation
- 10 kΩ Resistors – used for flex sensor voltage dividers
- Li-Po Battery + Boost Regulator – portable power supply
- USB Cable / OTG Cable – for communication and text-to-speech output
- Glove – wearable base for mounting sensors



Methodology

- Hand gestures are captured through sensors embedded in the glove
- The microcontroller analyzes the gesture patterns and identifies the intended sign
- Recognized gestures are translated into corresponding text
- The text is then converted into clear spoken output using a text-to-speech system
- The process happens in real time, allowing smooth and continuous communication

FEATURES

- Converts hand signs into speech
- User-friendly Model
- Real-time gesture recognition
- Customizable and can be used and trained for other purposes
- Can be trained for multiple sign languages
- Portable and cost-effective
- Can store model training data

OUTCOME

- Can be used for communication by speech-impaired individuals
- Improves accessibility and inclusivity
- Provides a foundation for further research in gesture vocalizer

FUTURE ENHANCEMENTS

- Add direct voice output from the glove itself
- Enable wireless communication for more freedom of movement
- Expand the gesture vocabulary to include more phrases and sentences
- Add personalisation so users can create their own custom gestures
- Develop a mobile app for seamless setup and customization.
- Integrate AI/ML for smarter and more accurate gesture recognition

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